ASCAT-DESTEKLI TIGGE RUZGAR HATA DUZELTMESI

Thiessen poligonlari ve XGBoost ile pilot dogrulama

Ozet

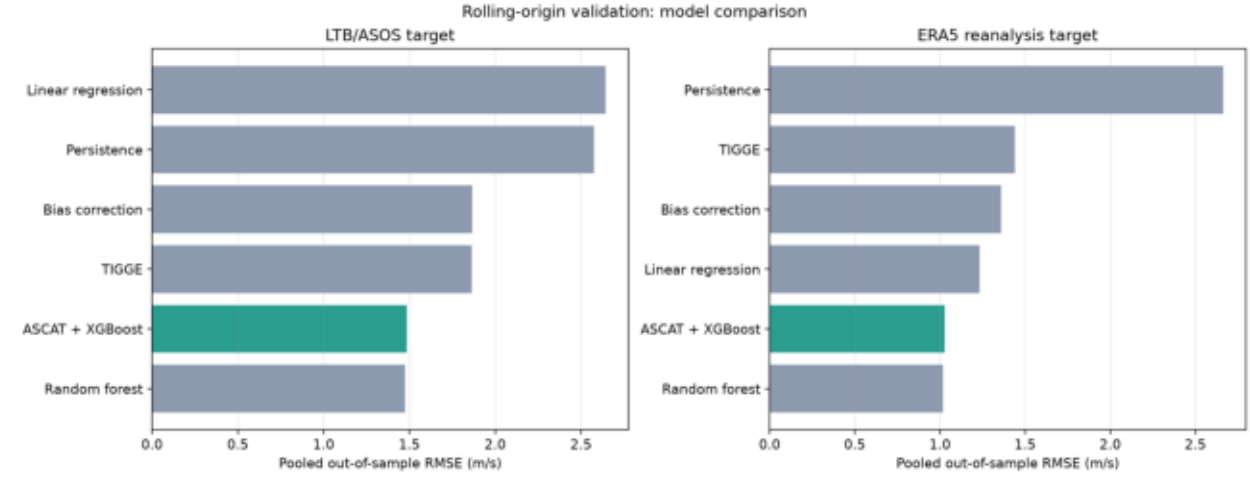
Bu rapor, 12 saatlik TIGGE ruzgar tahminlerini tahmin baslangicindaki ASCAT gozlemleriyle duz

Ana bulgu

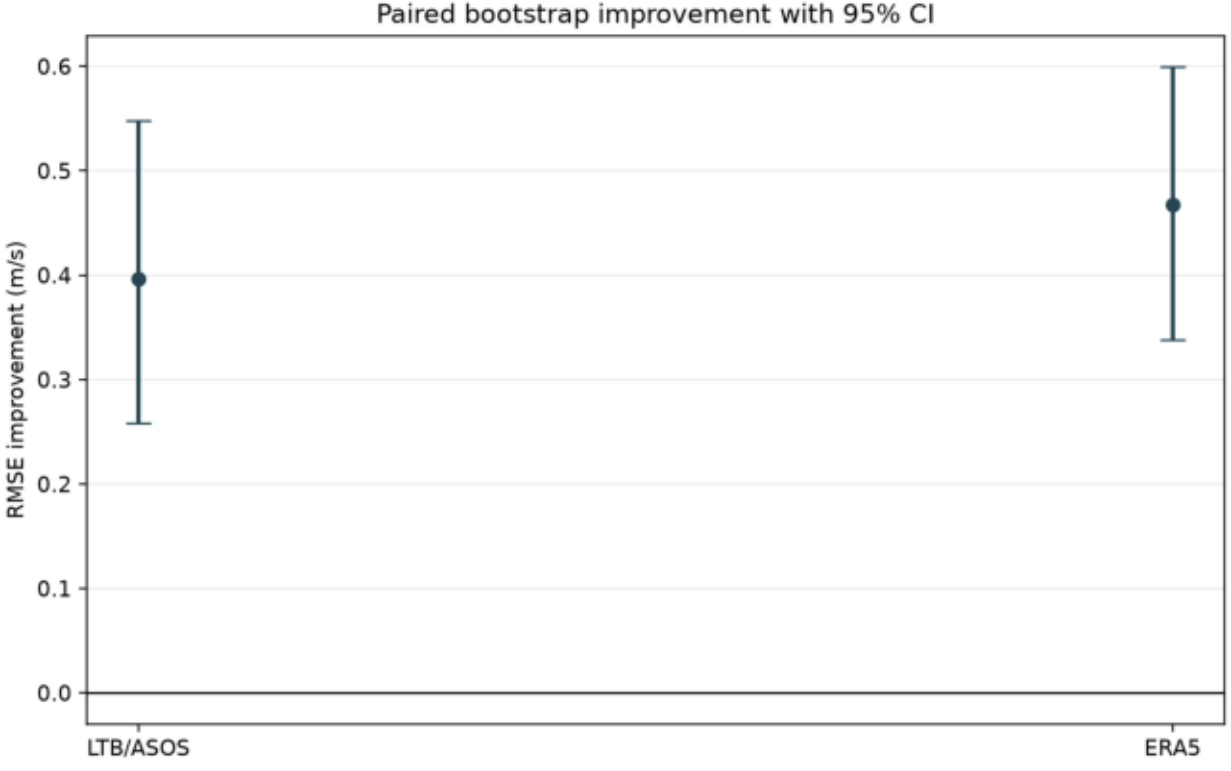
LTB hedefinde RMSE azalmasi: 20.37%\nERA5 hedefinde RMSE azalmasi: 28.54%\nKum

Sinir: Sonuclar Ocak-Nisan 2021 ve 113 forecast origin kumesine dayanan pilot calismadir.

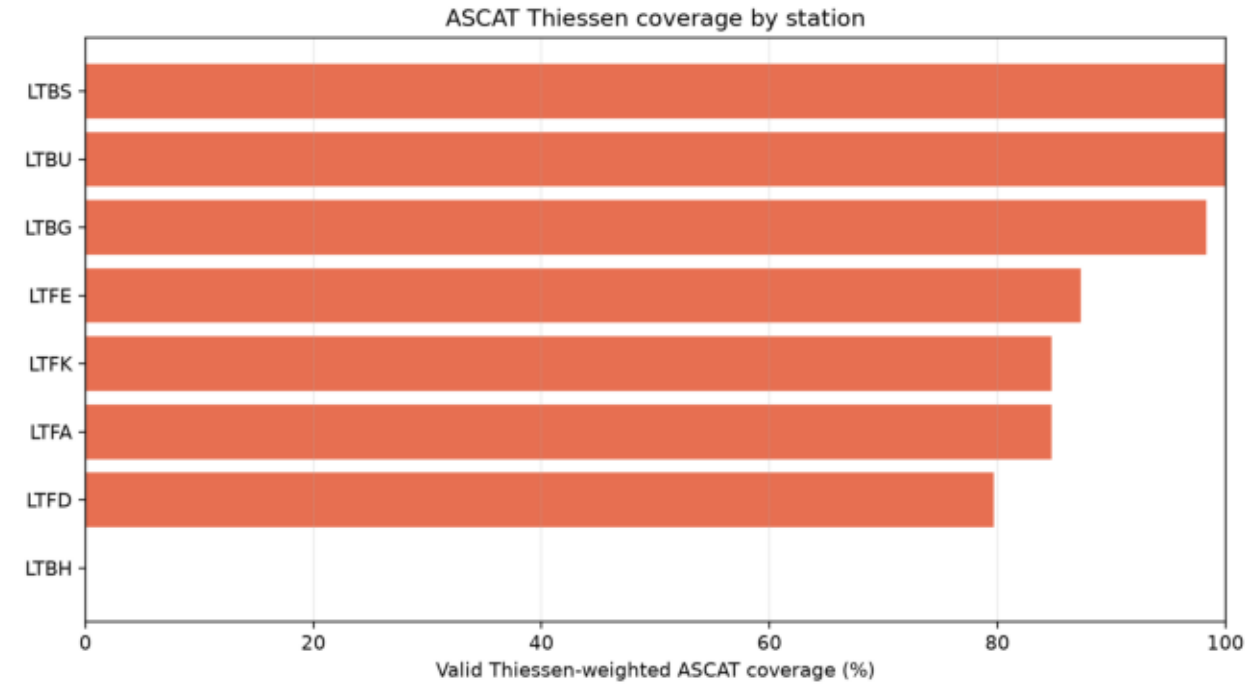
Model Rmse Comparison



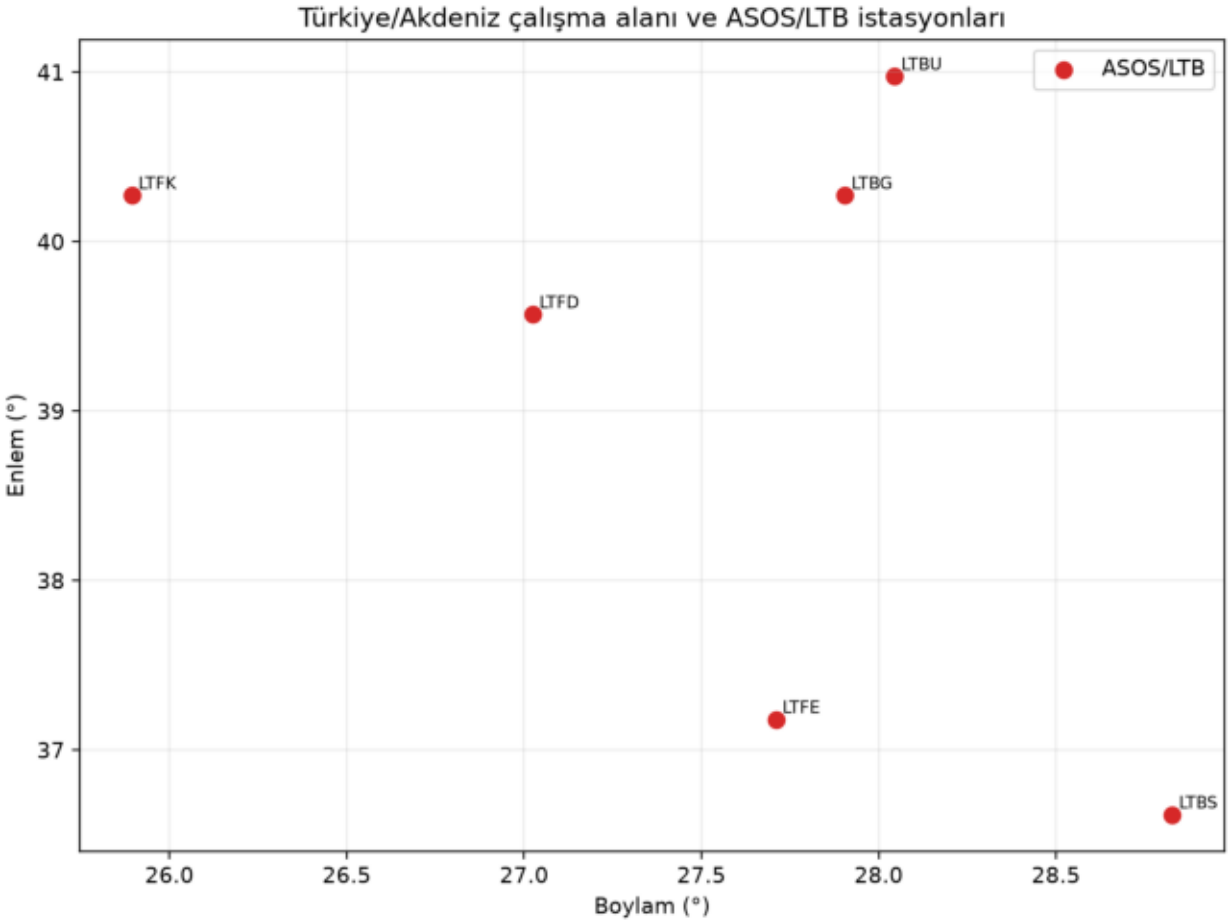
Rmse Bootstrap Improvement



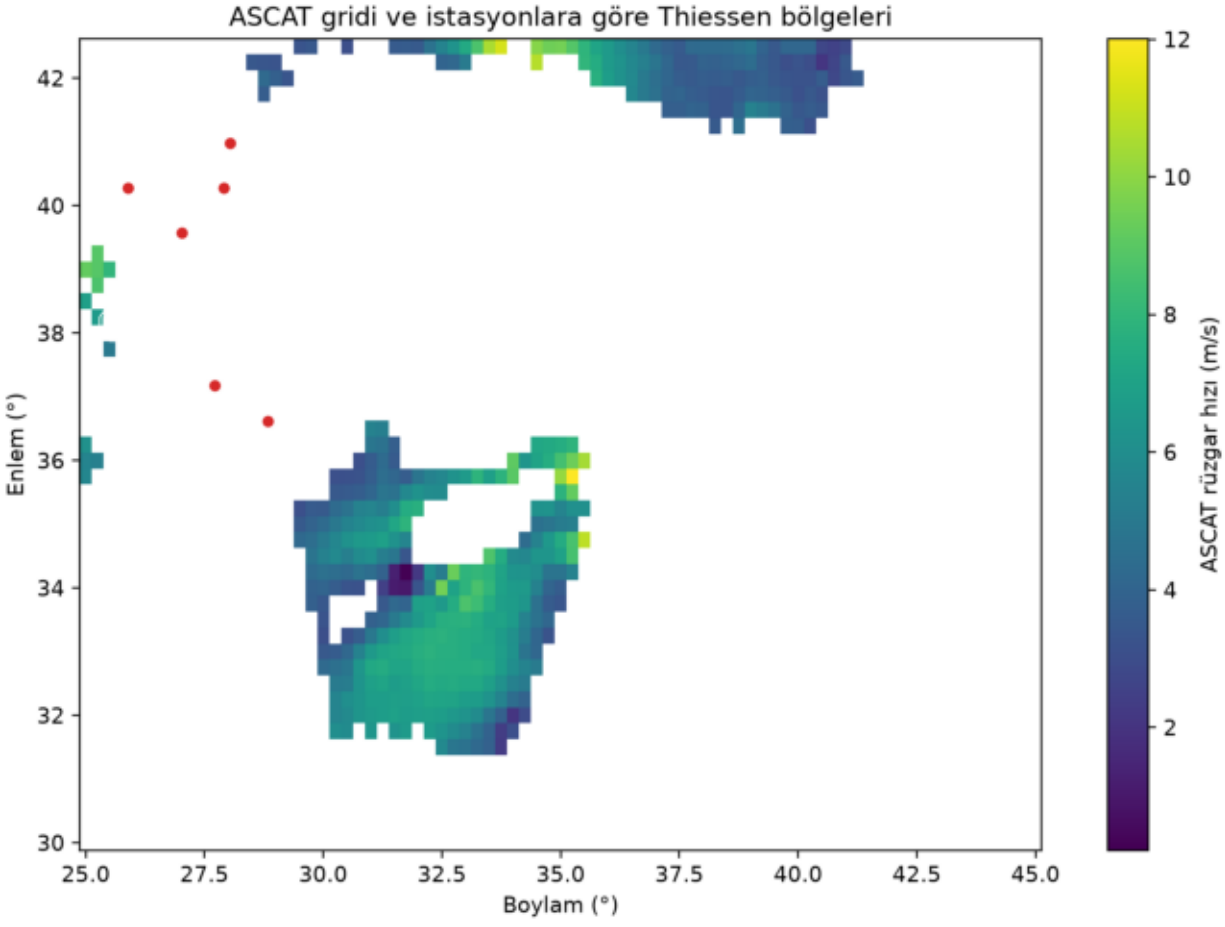
Thiessen Coverage



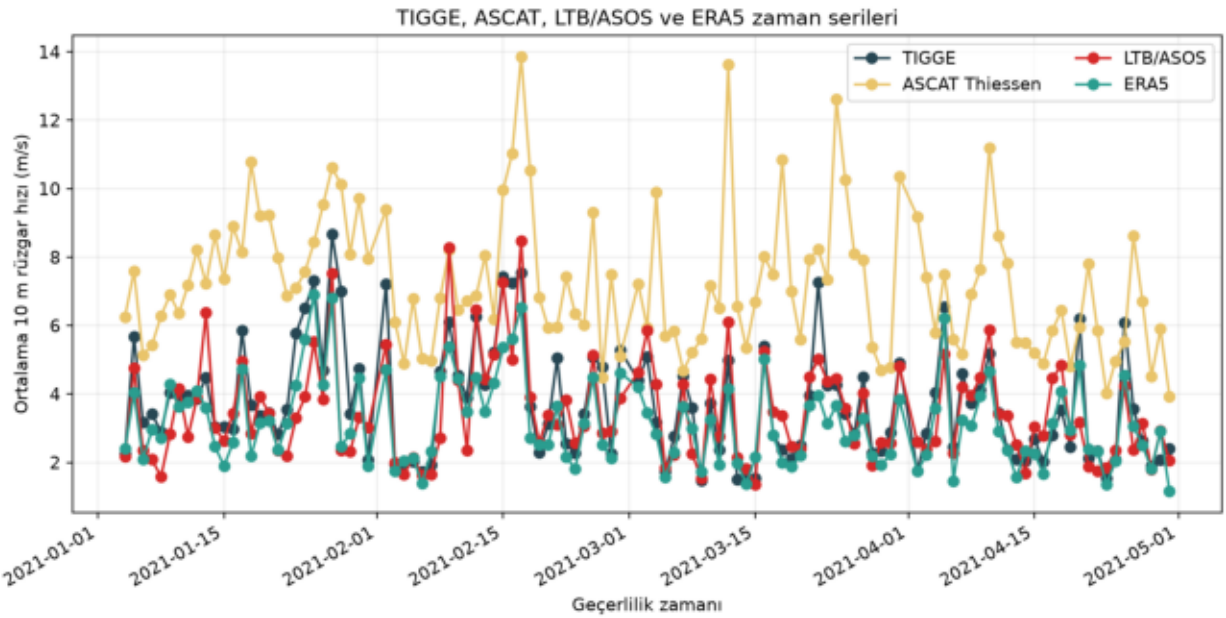
Study Area Asos Stations



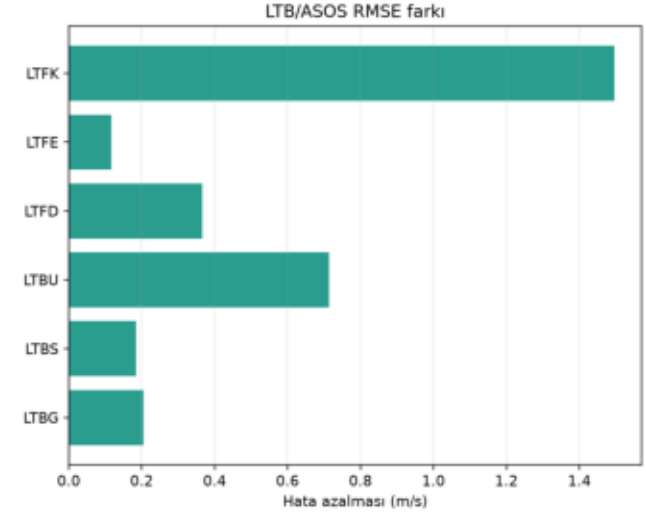
Ascat Grid Thiessen

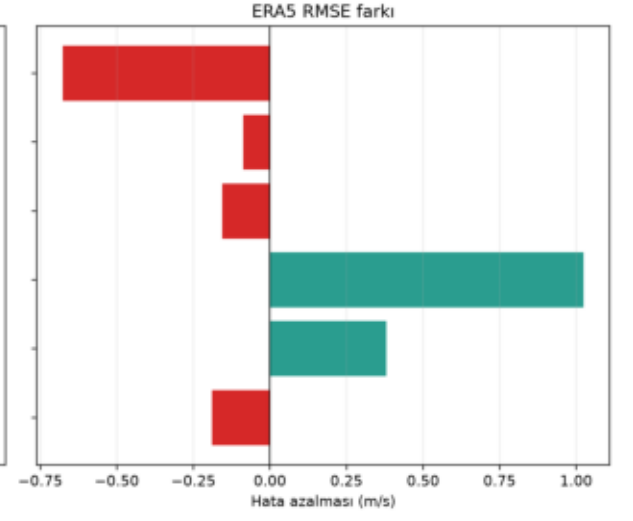


Wind Time Series

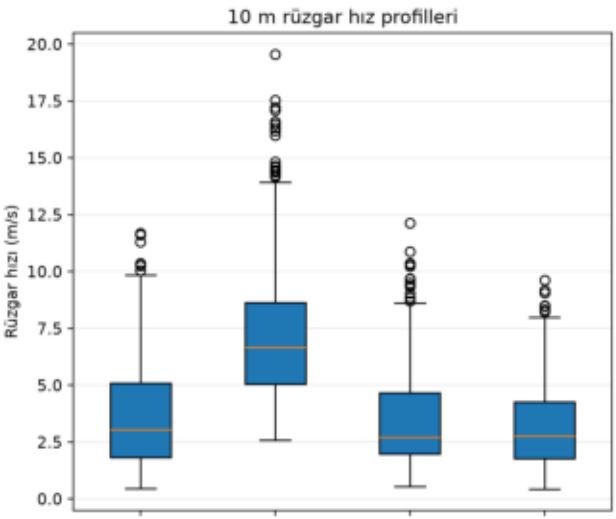


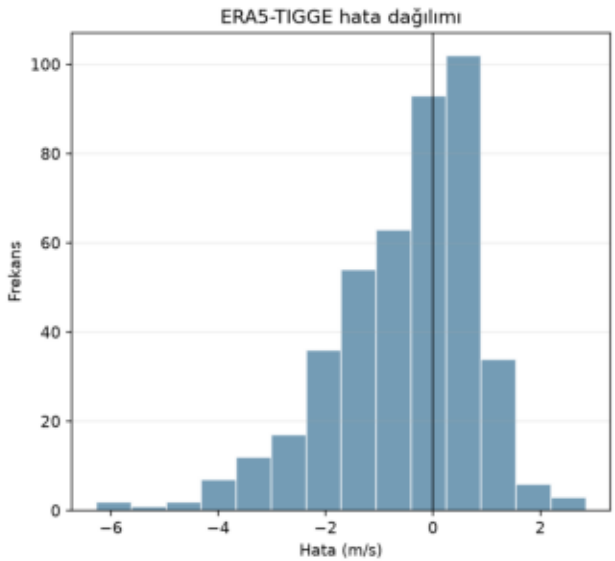
Station Rmse Difference



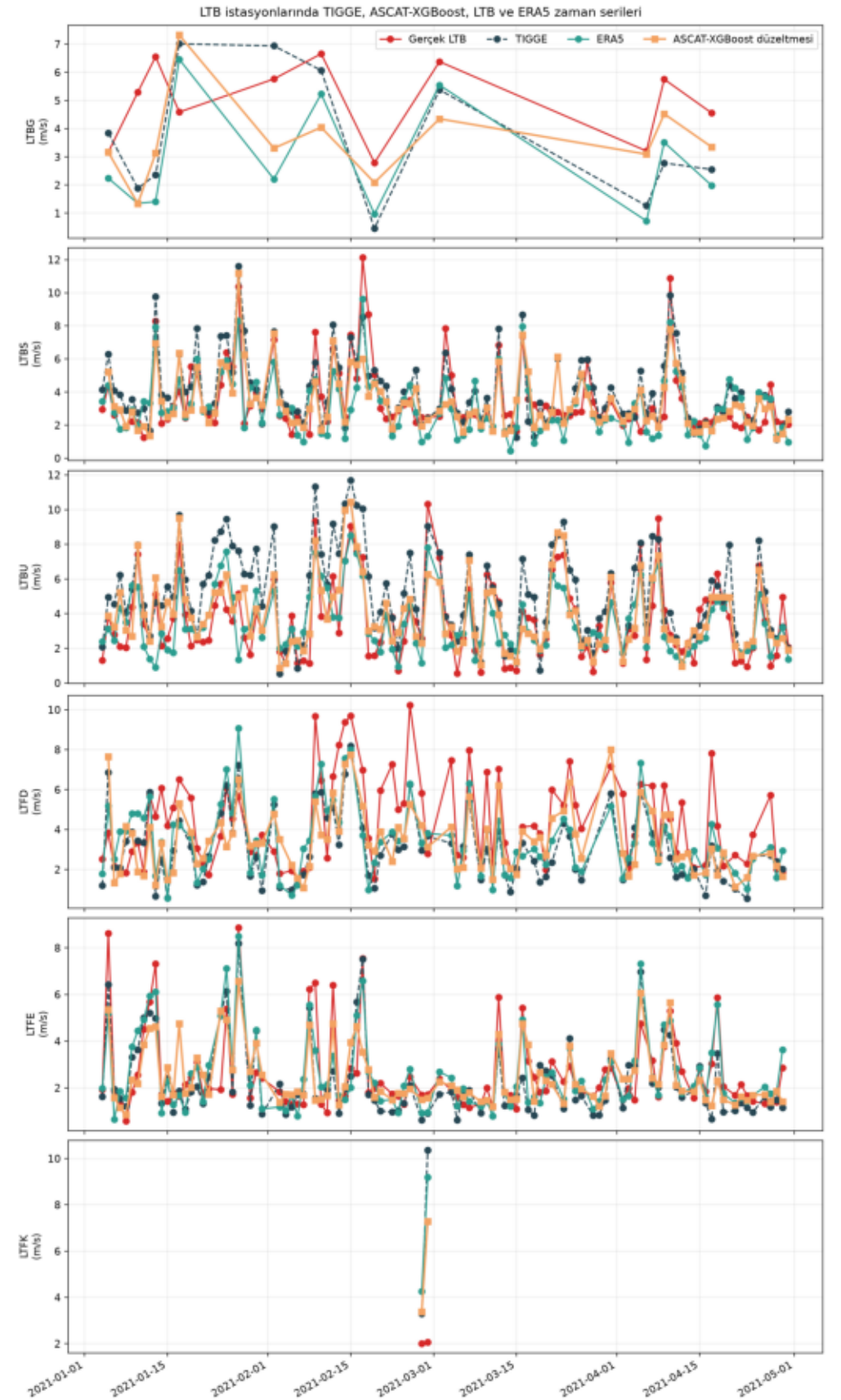


Wind Profiles Error Distributions

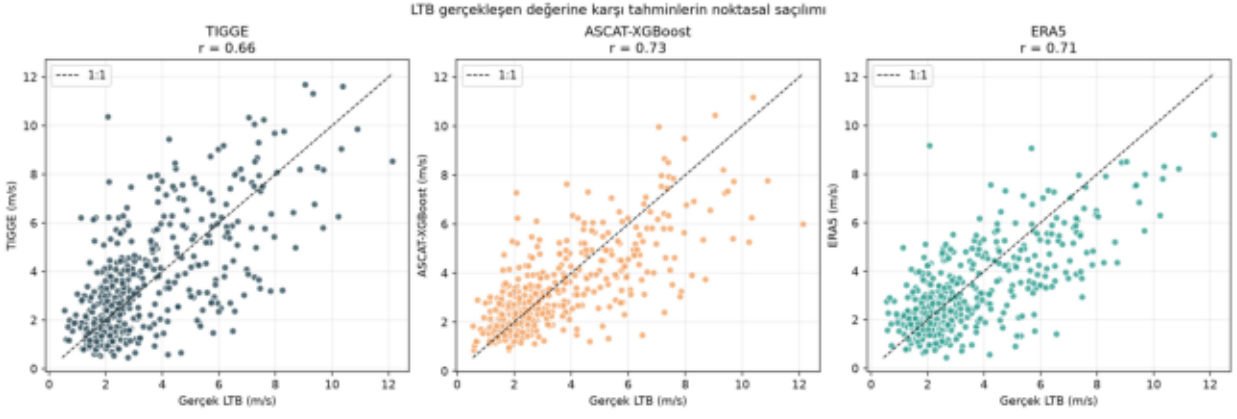




Ltb Forecast Timeseries



Ltb Forecast Scatter



Rolling-origin sonuclar

target	model	samples	mae	rmse	r2
ERA5	Persistence	428	2.056	2.663	-1.183
ERA5	TIGGE	428	1.056	1.439	0.363
ERA5	ASCAT Thiessen XGBoost	428	0.746	1.029	0.674
ERA5	linear regression	428	0.831	1.232	0.533
ERA5	random forest	428	0.743	1.016	0.682
ERA5	TIGGE bias correction	428	1.087	1.359	0.432
LTB	Persistence	428	1.825	2.575	-0.445
LTB	TIGGE	428	1.405	1.863	0.243
LTB	ASCAT Thiessen XGBoost	428	1.052	1.483	0.52
LTB	linear regression	428	1.225	2.643	-0.523
LTB	random forest	428	1.021	1.474	0.526
LTB	TIGGE bias correction	428	1.411	1.865	0.241

İstasyon-eşitlikli metrikler

target	model	equal_station_mae	equal_station_rmse
ERA5	TIGGE	1.037	1.301
ERA5	ASCAT Thiessen XGBoost	0.987	1.244
LTB	TIGGE	2.061	2.56
LTB	ASCAT Thiessen XGBoost	1.547	1.964

Formulasyonlar

$$U = \sqrt{u_{10}^2 + v_{10}^2}$$

$$e_t = U_t^{ERA5} - U_t^{TIGGE}$$

$$\hat{U}_t^{corr} = U_t^{TIGGE} + \hat{e}_t(X_t)$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{U}_i - U_i)^2}$$

$$z_i = (x_i - \bar{x})/s_x$$